Collector transition and exceptional-branch source audit

Source inspection only. Eight transition obligations and eight ordered dispatch regions. No scientific execution, new measurements, or Lean replay. Existing component results are mapped to remaining composition obligations; no theorem is promoted by this report.

Central invariant The next composition target is decoded state d = aP + bQ in the Jacobian class group. Canonicality and Mumford polynomial validity are separate representation facts. Both trial candidates may insert relations, while only the selected candidate advances the walk.

Source aliases collector: retained projective_state_collector.rs; law: retained projective_law_gate.rs; three_leads: current repository src/cryptanalysis/three_leads.rs. Complete paths and hashes are listed at the end and in status.json.

Collector transitions

1. transfer

Source: collector:132-161

Existing evidence: normalization-input-guards, normalization-inverse-support, monic-remainder-zero-test; new residual Acceptance reported by parent

Remaining composition: Bind actual source transfer to class homomorphism, nonzero normalized leading scale, canonical words and nonzero weights. Existing j.valid checks prove representation acceptance, not class-image equality.

2. initialization and restart

Source: collector:173-198,221-228,344-352

Existing evidence: word-add11-validity-review, word-doubling-review, mixed-dispatch-word-review, sameu-word-review

Remaining composition: Prove scalar loop and add dispatch preserve class labels; initialize d=aP+bQ and each jump=A_iP+B_iQ. Preserve on both restart sites.

3. candidate transition

Source: collector:242-252

Existing evidence: same component addition proofs

Remaining composition: For each of TWO candidates derive nd=d+jump and na=a+A,nb=b+B modulo r. Relation insertion can happen for both candidates although only best advances. Needs class-sum semantics and [r]P=[r]Q=0.

4. identity return

Source: collector:253-257

Existing evidence: identity representation and word canonicality components

Remaining composition: Prove zero class implies na+nb*k=0 mod r via injective generator scalar map; justify prime r or unit nb before inverse.

5. split normalization

Source: collector:258-275; law:582-589

Existing evidence: weighted decode and normalization packets

Remaining composition: Prove split_possible discriminant scaling reflection for nonzero weights; normalized u roots and repeated-root multiplicity; curve validity preserved. Native asserts ordinate equation only after roots, not complete j.valid here.

6. signed physical relation

Source: collector:288-320

Existing evidence: relation-adapter-review, negation and Mumford validity components

Remaining composition: Bind effective Mumford divisor to signed point classes, including repeated roots and y=0. Canonical min with negation is only duplicate key; na,nb retain original sign. Define canonical node ordinate and identity anchor0.

7. graph insertion

Source: collector:321-334; three_leads:1558-1712

Existing evidence: relation-adapter-review and target-information review scopes

Remaining composition: Prove find potentials/path compression, union-by-size reversal, fixed anchors, same-root positive/negative cycles and constraint extraction. Need homomorphism h:J->Z/r with h(P)=1,h(Q)=k; nodes need not individually be r-torsion if h exists.

8. continuation selection

Source: collector:337-354

Existing evidence: no complete induction found in inspected packets

Remaining composition: Commit selected candidate together with its matching na,nb; graph remains sound after relations from unselected candidate; duplicate-induced restart resets labels coherently.

Ordered dispatch regions

1. identity and1+1

Source: law:612-615

Existing evidence: word-add11-validity-review; word-doubling-review

Remaining composition: Complete ordered inverse/equal/distinct assembly and class-sum semantics, rather than redo component validity.

2. 1+2 /2+1

Source: law:616-622,828-878

Existing evidence: mixed-dispatch-word-review

Remaining composition: Reviewed source-order canonical validity and None exclusions available under original premises; supply reachable-state premises and class-sum theorem.

3. same-u inverse

Source: law:925-927

Existing evidence: sameu guard/word packets

Remaining composition: Inverse precedence and identity class sum; do not equate arbitrary record equality with class equality.

4. same-u equal/double22

Source: law:928-930,727-826

Existing evidence: degree22-guards-review and bzero producer packets

Remaining composition: Ordered bzero/szero identity exclusion established; bzero support geometry does not prove valuations/class sum. Generic double22 and all output branches still require composed validity/class/weight closure.

5. same-u mixed

Source: law:931-951

Existing evidence: sameu-composition-review, sameu-word-review

Remaining composition: Zero-slope exclusion, extraction/doubling validity established with global smoothness; supply constructor smoothness binding and class sum.

6. distinct support B=0

Source: law:958-999

Existing evidence: degree22 guards; mixed dispatcher components

Remaining composition: shared_zero_q_delta exclusion available. Need original-input proof of extracted p1..p4 validity/nonzero weights; mismatch exclusion; sequential p2+2p1 then p4 addition. Both explicit first secondary None and returned second None need exclusion.

7. B != 0 C=0

Source: law:1000-1023

Existing evidence: no full composed result identified in inspected packets

Remaining composition: Prove literal degree-one payload, canonicality/nonzero weights, normalized validity and class sum.

8. B != 0 C != 0

Source: law:1025;636-726

Existing evidence: common formula source present

Remaining composition: Bind common helper guards to outer values and prove payload/canonicality/weights/validity/class sum.

Interpretation and boundaries

A graph-label proof may use a homomorphism h from the Jacobian group into Z/r with h(P)=1 and h(Q)=k. Individual factor points need not each be r-torsion if that homomorphism is established. Its existence and normalization remain proof obligations. Native expect calls make mixed/degree-two None exclusions concrete panic-freedom conditions. Raw projective equality governs scheduling and deduplication; it must not silently mean canonical class equality.

Existing independent review statements inspected as evidence descriptions; no new hash/replay audit of all their dependencies. Absence claims limited to inspected packet set.

Bound source and evidence files

/home/ec2-user/crypto-pr2304-persist/research/projective_law_followup_20261022/src/bin/projective_state_collector.rs

96246448e56f46b57e0ddd751990cf11fab122fbc04355b3e0ec6530194f1e35

/home/ec2-user/crypto-pr2304-persist/research/projective_law_followup_20261022/src/bin/projective_law_gate.rs

5976e7f30b3e9c5c49a5fc0f4095404b7322832e4b02f467bc3cb7e0d9dcc8c0

/home/ec2-user/crypto-pencil-signed-graph-20261029/src/cryptanalysis/three_leads.rs

26e99b56dace8f848d055e8709bb989622e4caea65aec26d63e00805153dc3c6

/var/tmp/pencil-portfolio-audit-20261101/matrix.json

b69bf55ca42edc7144a6d7330ecc9ccdc0a8ed4fcde228be0fb16f6ff4efa7ff

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/mixed-dispatch-word-review/REVIEW.md

283159612d65179db91163b508e16d00e726cbc4328127516f8412ed8e7f90ec

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/sameu-composition-review/REVIEW.md

344ccf4c551ef48e735bcf389836c56054c671f9d9c3f0ae6b6759179d83235c

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/sameu-word-review/REVIEW.md

2b655991fc2aa14514338df1a63aa25fdc77461cbabd978088d76696fcb46e40

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/word-doubling-review/REPORT.md

d6c042440314ae1666e009a5356a1e3a460aa81ca0f29feb90269f82df550e6e

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/word-add11-validity-review/REVIEW.md

d1198ac53a7f3822711ac397c3e93ef6a259aec72a71988f104e43e21caf08f3

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/degree22-guards-review/REVIEW.md

5a8152296da083c95d6ec701eeb0a2ddf71fc2fe0ffce180c6c0f615111973e4

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/degree22-guards/REPORT.md

ab4b5c1ea780bff81459d1c474c0e459f63e6cde32e3a5a26b2a68e05799be89

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/degree22-bzero-word-producer/status.json

a455bf3eb3366d9a641c6c999012aa4489eca186be042c5fcee2a7df537506b0

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets/degree22-bzero-geometry-producer/status.json

adff780b7ecb0832557847208b8398370ea9a98ba67c3f95dd640ae5edd0afeb